

Since the spring of this year, I began to study quantum mechanics, read literature on physics, and study programming more deeply.

The discussion about describing scientific measurements and comparing the different quantitative characteristics that define them is very interesting.

This is probably a matter of opinion, so it's probably important to just state your opinion.

Physics is an experimental science. Experiment is a source of knowledge and a criterion for the truth of any hypotheses and theories. Currently, physical methods of research and control are widely used in all branches of technology without exception. Therefore, it is very important, during the learning process, to master a wide range of physical techniques, gain experience in using various measuring instruments and, no less important, learn to correctly assess the errors that inevitably arise.

The knowledge, skills and abilities that should allow solving these problems are intended to be given to the student by working in the laboratory. Performing laboratory work is in many ways similar to conducting a scientific experiment, has the same goals and consists of similar steps.

Many people around the world who devote themselves to experimental work experience great pleasure not only from the results obtained, but also from the fascination of the research process itself. And although in the educational laboratory the main goal of the experiment is to confirm already known patterns, work in it, nevertheless, should be considered as a small but exciting journey into the world of science.

Measurement is the process of experimentally obtaining one or more values of a quantity that can be reasonably assigned to the quantity.

Measurement is finding the value of a physical quantity experimentally using special technical means

The essence of measurement is the comparison of the measured value with a known value taken as a unit, that is, a standard. To measure any physical quantity Q means to compare it with another quantity q , taken as a unit of measurement, and express the first in fractions of the latter.

The object of measurement is any selected characteristic of a certain object or phenomenon that makes up a fragment of reality. This characteristic manifests itself in the objects under study or in one object with different intensities, which is expressed by different states of the characteristic,

A person, a subject of cognition, implementing a cognitive act, decides how detailed the cognition should be and what goal it pursues.

The result of measuring a quantity is a set of quantity values assigned to the measured quantity along with any other available and relevant information.

The measurement result can be expressed as follows: $Q = q [Q]$, where Q is the actual value of the quantity found during the measurement; q – numerical value of a quantity in accepted units, $[Q]$ – unit of quantity.

Experimental studies of some object or phenomenon, testing of industrial products, control of mechanisms or processes are unthinkable without measurements of physical quantities characterizing the state of the object.

Therefore, any measuring, testing or control system includes primary measuring transducers - sensors - as the most important components. The main function of sensors is to convert the values of various physical quantities into electrical quantities - voltage, current, time-frequency parameter of the electrical signal.

In the list of problems that need to be solved when developing and using measuring systems, one of the most important is that it is necessary to measure not one, but various physical quantities - for example, pressure, vibration, rotation speed, deformation and much more, using different principles actions, characteristics and design of sensors.

An analysis of the problems of obtaining information using sensors, carried out by specialists for many years, showed that these problems need to be solved in a comprehensive manner, taking into account the features of integrating sensors into systems, taking into account the prospects for the development of microelectronics, circuitry, microprocessor technology and other factors.

And this is possible if the sensors perform not only their main function - the transformation of physical quantities, but also a number of functions that are implemented by subsequent units of the measuring system, as well as some special functions

The result is the adoption of the IEEE 145 family of standards.

Byte order

All modern computers address memory byte by byte. This means that any memory cell can be addressed or accessed directly. So, for example, a 32-bit word consists of four bytes, let's denote them as $B_0 B_1 B_2 B_3$. The address of byte B_3 is the address of byte $B_0 + 3$. Now suppose that we store a number in single-precision IEEE format in such a 32-bit word. The IEEE number bits will be stored there as $\sigma a_1 a_2 \dots a_8 b_1 b_2 b_3 \dots b_{23}$, where σ is the sign bit.

The first 8 bits of this number, $\sigma a_1 a_2 a_3 a_4 a_5 a_6 a_7$, constitute one byte. Where is this byte stored - in place B_0 or in place B_3 ? The answer to this question depends on the byte order adopted for a given computer architecture. So, for example, on x86 this byte will be stored in B_0 , and on SPARC in B_3 . The byte order of IEEE numbers is usually the same as the byte order of your computer. When exchanging numbers in IEEE format, for example, over a network, it should be taken into account that the byte order on the sending and receiving sides is not necessarily the same.

Intelligent monitoring systems and intelligent environments:

The system core of the intellectual environment is expressed in the form: a distributed system for perceiving the parameters of the physical and technical environment + an intelligent system + a distributed system for influencing the physical and technical environment. An important place is occupied by information coming from sensors, its analysis and interpretation for identifying knowledge.

Those. a full-fledged process of intelligent monitoring in the form of a unity of processes: measurement - evaluation - reasoning - calculations. The unity of these processes is one of the system principles on which the concept of cognitive measurements is based.

Systemic principles of cognitive measurements:

Measurements that extract and use metrologically validated knowledge using Bayesian intelligence.

An extension is proposed - cognitive measurement should be understood, inter alia, as the organization of measurement as an automated cognitive process aimed at ensuring computer understanding and pragmatic interpretation of measurement results.

This concept involves the construction of cognitive information-measuring devices that provide granulation of measurement information, and is based on the following system principles: 1) the principle of openness of measurement as a cognitive process; 2) the principle of unity of measurements, assessments and reasoning; 3) the principle of synthesis of theories of truth when interpreting measurement results; 4) the principle of granulation of measurement information.

Openness of measurements means their dependence on goals, measurement environment, measurement tools, and finally, inclusion in more general processes.

In accordance with the principle of unity of measurements, assessments and reasoning, measurement is understood as a cognitive process associated with the construction of a granular structure of information in the form of a two-level hierarchy: 1) at the lower level, regular measurements are carried out using a set of sensors, the results of which provide fine-grained information, 2) At the top level, the obtained results are mapped onto a pragmatic scale of aggregated assessments - axiological or deontic - expressing coarse-grained information (the obtained values of the measured parameter are “normal”, “almost normal”, “not normal”, etc., for example, measured wind speed value is “almost normal”).

We build an analysis of the connection between the result of measurement (as a process of knowledge formation) and understanding on the basis of two concepts of truth: Tarski's theory of correspondence and Peirce's pragmatic theory of truth. Among the unconditional advantages of Tarski's theory, which separate it from all its predecessors, we can point out the following:

1. A. Tarski's theory is a formal, i.e., mathematical (or quasi-mathematical) theory.
2. It offers a detailed, precise and rigorous definition of truth.
3. In the theory of A. Tarski, a serious difficulty for all theories of truth is overcome - semantic paradoxes, in particular, a solution to the Liar paradox is given.
4. A. Tarski's theory makes a serious contribution to the development of modern logic and scientific methodology.
5. A. Tarski's theory is relatively independent of traditional philosophical disputes.
6. It raises a range of new philosophical questions and offers new approaches to their solutions

At the same time, C. S. Peirce is rightfully considered one of the most original and multifaceted philosophers that has ever been produced to the light of America. As an innovative intellectual, he anticipated the ways development of a variety of scientific disciplines. His research left a noticeable mark both in the exact and natural sciences, as well as in the humanities.

He was a mathematician, astronomer, chemist, surveyor, cartographer and an engineer, but also a psychologist, philologist and historian of science, was one of the first in the United States to engage in experimental psychology and the first to use the wavelength of light as a measurement measure. His posthumous fame was achieved by works in the field of logic and semiotics (the science of signs to convey information), but he was also the author original metaphysical system.

Into the history of philosophy C. S. Peirce entered as the founder of the philosophy of pragmatism, another direction of his intellectual creativity .

In the article "What is pragmatism ?" (1905) Peirce wrote that "perhaps the most striking feature of the new theory was the recognition of an inseparable connection between rational knowledge and rational purpose."

"Consolidation of Beliefs." In this article, Peirce introduces two important concepts - the concept of "doubt" and the concept of "belief" - a polysemantic English word that can be translated as belief, opinion, belief. In Peirce, of course, we are not talking about a religious sense, but about the psychological states of faith and doubt that everyone experiences.

Both doubt and faith have a positive effect, so each is necessary in its own way. Faith does not force us to act immediately, but under certain circumstances it forces us to act in a certain way, that is, it acts as a predisposition to action. Doubt immediately stimulates us to action until we overcome it.

Thus, all human activity has the structure of a transition from doubt to faith. The very transition from doubt to faith - any, and not just those related to scientific activity - Peirce calls "research".

Peirce's doubt arises naturally and is connected first only with the appearance of circumstances that do not fit into the usual picture of the world. John Dewey later introduced the concept of "problematic situation" and formulated the main features of such a situation, however the original idea belonged, as we see, to Peirce.

Thought is one of the systems of relations between sensations. Despite the fact that the essence of thought is movement, the main goal of any mental activity is to achieve peace, i.e., to achieve belief. One does not contradict the other, since, "being a place where thought stops, belief is also an area that draws thought into a new movement."

On the other hand, belief itself is characterized by Peirce through three functional properties: 1) awareness; 2) eliminating irritation caused by doubt; 3) establishing a habit. Behind our thoughts, even the most abstract ones, lies a system of sensations.

The pragmatist maxim, which became known as "Peirce's principle," goes as follows: "...consider what kind of consequences, which can be of practical importance, we believe the object of our concept to have. Then our concept of these consequences is the complete concept of the object."

The real is "that whose properties are independent of what anyone can think about them," that is, something independent of our thinking. The concept of reality, according to this rule, is reduced to the "tangible consequences" that real things cause. Therefore, we need to find out what these consequences are.

The main effect (perceived effect) of real things is the production of beliefs. Peirce believes that true reality consists in the opinion of it with which everyone agrees. (Does this happen?).

What is pragmatism?

We will focus on several ideas that are fundamental to understanding the essence of pragmatism. If pragmatism is a philosophy, then the question of what should be considered the beginning of philosophy comes up on the agenda. Pragmatism is engaged in "purification", a procedure that much later, with the light hand of L. Wittgenstein, received the name of philosophical "therapy".

We are talking about cleansing philosophy from "ontological metaphysics," which Peirce, without mincing words, calls "meaningless gibberish" and "outright absurdity." What do we get as a result? As Peirce writes, "a philosophy cleansed of such rubbish will remain only a number of problems that are quite possible to investigate using the methods of observation characteristic of genuine sciences." The above statement is reminiscent of similar statements made by positivists.

But Peirce does not object to such a parallel and even calls pragmatism a teaching close to positivism. What distinguishes pragmatism from positivism is the following: 1. preservation of

purified philosophy; 2. full acceptance of the basic body of our instinctive beliefs; 3. persistent adherence to the truth of scholastic realism.

The concept of truth in pragmatism.

Representatives of pragmatism in general and Peirce in particular have never been fans of the concept of “truth.” It is not the truth that is important, but a strong opinion or belief. When we have reached a firm and unambiguous opinion on this or that matter, we are no longer interested in its truth or falsity. Peirce has a negative attitude towards the coherent (interconnected) concept of truth, the most widespread among the philosopher’s contemporaries. And, of course, Peirce is an opponent of the metaphysical absolutization of truth.

The fundamental role of doubt in the process of cognition forces us to rethink the concept of truth. In this case, it is necessary to distinguish between two concepts of truth: 1) belief, leading to behavior that satisfies the corresponding desire, and is useful for the survival and adaptation of a person; 2) the final conviction of the majority as a natural and necessary result of a long study conducted using the scientific method.

The later Peirce further narrows the concept of truth, limiting it to what is accepted as true in science. The concept of truth is now characterized by fallibilism (fallible, fallible), according to which the idea of “scientific truth” also includes the extent to which it is false.

One of the negative consequences of fallibilism is the elimination from science of the concept of absolute certainty, the basis and engine of scientific progress. Peirce, deeply devoted to science all his life, finds a way out of this situation through the concept of “practical certainty.” What justifies itself in practice is practically infallible and practically reliable.

Deduction, induction and abduction.

Any knowledge, according to Peirce, must come from facts and be confirmed by observation. As for the “types of reasoning” that lead scientific research to some positive result, i.e. knowledge, there are three of them: deduction, induction and abduction (retroduction).

Peirce's innovation, of course, is abduction, which complements the two well-known methods - deduction and induction, which, as previous philosophy believed, exhaust our capabilities in the field of knowledge.

The abduction method is responsible for the birth of a hypothesis and in this sense is of paramount importance. According to Peirce, abduction involves two stages: generating explanatory hypotheses and selecting from among these hypotheses the most promising explanation for the phenomenon under consideration.

A significant property of abduction is its riskiness. Indeed, choosing from a range of hypotheses, even if subject to experimental testing, is a risky undertaking, since it determines the further course of the research. The justification of the choice will become clear only in the end.

In this regard, Peirce speaks of the need for an investigative “feeling” or “instinct,” since “a person who does not have an inclination that is consistent with the inclination of nature itself has not the slightest chance of understanding nature.”

Just as the decomposition of the number 3.14 in the form of a decimal fraction can only be more accurate, but can never be completed, Peirce believes, so our knowledge can only become more and more true, but never in a finite number of steps is able to achieve absolute truth. According to Peirce, this is impossible for the reason that absolute truth is knowledge of all infinite reality.

But for practical matters such knowledge is not necessary. It is necessary as an ideal limit, or better to say, as a hidden tendency that inexorably directs research, despite individual deviations, towards the truth. Without the existence of such a limit, no finite number of trials could distinguish a true theory from a false one (let the reader try, based on three or four tosses, to determine the probability of getting the emblem of an asymmetrical coin).

The existence of a limit and its functions in scientific knowledge are determined by the nature of reality itself, which expresses its necessary existence in the form of laws. As an objective regularity, the law of nature serves as an ideal limit that governs the process of creating more and more accurate, but never completed theories.

In other words, according to Peirce, the process of self-correction of scientific knowledge has objective and subjective grounds - in the nature of reality itself and the characteristics of our knowledge. Neither one nor the other is in doubt.

The concept of the scientific method created by him, in which all its parts - abduction, deduction and induction - mutually determine and correct each other, represents a real discovery, significantly ahead of similar developments of the 20th century.

